THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The value of y in $13x + 5y = 67,\ 6x + 7y = 45$ is: 1

(a) 7 **(b)** 4 **(c)** 3 **(d)** -3

2. The product of the roots of $3x^2 + 7x - 6 = 0$ is: 1

(a) 2 **(b)** $\frac{7}{3}$ **(c)** -2 **(d)** $-\frac{7}{3}$

3. The number of terms in the AP $8, 11, \ldots, 98$ is: 1

(a) 30 **(b)** 32 **(c)** 31 **(d)** 33

4. The areas of two similar triangles are 36 cm² and 441 cm². The ratio of their perimeters is 2: 1

(a) 9 **(b)** 14 **(c)** 7 **(d)** 49

5. The point which divides $(8, 0)$ and $(0, 4)$ in the ratio 1:1 has coordinates summing to: 1

(a) 6 **(b)** 12 **(c)** 16 **(d)** 3

6. If $\sin\theta = \frac{12}{13}$, then $\sec\theta + \tan\theta =$ 1

(a) 5 **(b)** $\frac{25}{12}$ **(c)** $\frac{13}{5}$ **(d)** $\frac{1}{5}$

7. If the angle of elevation of the top of a tower is 30° at a point P and it is 60° at a point nearer to the tower, then the point nearer to the tower is at: 1

(a) the same distance as P

(b) half the distance of P from the foot

(c) twice the distance of P from the foot

(d) one-third of the distance of P from the foot

8. The length of a tangent from a point P to a circle is 12 cm and the radius is 5 cm. The distance of P from the centre is: 1

(a) $\sqrt{119}$ cm **(b)** 7 cm **(c)** 17 cm **(d)** 13 cm

9. A quadrant of a circle of radius 14 cm is cut from a square of side 14 cm, with the centre of the quadrant at one corner of the square. The area of the remaining part of the square is: 1

(a) 84 cm² **(b)** 42 cm² **(c)** 154 cm² **(d)** 196 cm²

10. A solid metal cuboid 24 cm by 12 cm by 18 cm is melted and recast into cubes of edge 3 cm. The number of cubes obtained is: 1

(a) 191 **(b)** 384 **(c)** 576 **(d)** 192

11. For the data with classes 80–90, 90–100, 100–110, 110–120, 120–130, 130–140 with frequencies 3, 6, 13, 15, 7, 6 respectively, The mean of the data is: 1

(a) 111 **(b)** 115 **(c)** 113 **(d)** 112

12. A red die and a blue die are rolled together. The probability that the first number is greater than the second is: 1

(a) $\frac{5}{12}$ **(b)** $\frac{7}{18}$ **(c)** $\frac{4}{9}$ **(d)** $\frac{7}{12}$

13. The largest number that divides 74 and 130 leaving remainders 2 and 4 respectively is: 1

(a) 9 **(b)** 36 **(c)** 72 **(d)** 18

14. If $x = 3$ is a zero of $x^2 - 8x + k$, then k is: 1

(a) 5 **(b)** 15 **(c)** 33 **(d)** 24

15. The pair $3x + ky = 9$, $6x + 6y = 18$ has infinitely many solutions when $k =$ 1

(a) 4 **(b)** 9 **(c)** 6 **(d)** 3

16. The roots of $x^2 - 2x - 15 = 0$ differ by: 1

(a) 15 **(b)** 8 **(c)** 9 **(d)** 2

17. The sum of all two-digit multiples of 4 is: 1

(a) 1200 **(b)** 1188 **(c)** 1092 **(d)** 1288

18. The altitude to the hypotenuse of a right triangle divides it into 3 cm and 12 cm. The altitude (cm) is: 1

(a) 6 **(b)** 15 **(c)** 36 **(d)** $\frac{15}{2}$

19. The centroid of the triangle with vertices $(0, 0)$, $(6, 3)$, $(3, 9)$ has x-coordinate: 1

(a) 3 **(b)** 9 **(c)** $\frac{9}{2}$ **(d)** 4

20. If $\cos\theta = \frac{5}{13}$, then $\csc\theta - \cot\theta =$ 1

(a) $\frac{2}{3}$ **(b)** $\frac{13}{12}$ **(c)** $\frac{1}{5}$ **(d)** $\frac{3}{2}$

21. A boy whose eyes are 5 m above the ground stands 8 m away from a vertical building. The angle of elevation of the top of the building from his eyes is 60°. The height of the building is: 1

(a) $6 + 8\sqrt{3}$ m **(b)** 13 m **(c)** $8\sqrt{3}$ m **(d)** $5 + 8\sqrt{3}$ m

22. The length of the tangent from an external point P to a circle is 21 cm and P is 35 cm from the centre. The radius of the circle is: 1

(a) 14 cm **(b)** 28 cm **(c)** 56 cm **(d)** 22 cm

23. A square is inscribed in a circle of radius 14 cm. The area of the circle outside the square is: 1

(a) 224 cm² **(b)** 616 cm² **(c)** 420 cm² **(d)** 392 cm²

24. A tent is a cylinder of radius 3.5 m and height 10 m surmounted by a cone of the same radius and height 12 m. The area of canvas (in m^2) needed to make the tent is: 1

(a) 357.5 m² **(b)** 352 m² **(c)** 220 m² **(d)** 137.5 m²

25. For the data with classes 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 12, 9, 5, 14, 5, 5 respectively, The median of the data is: 1

(a) 36.2 **(b)** 40 **(c)** 38 **(d)** 30

26. A coin is tossed three times. The probability of getting a head on the first toss is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{3}{8}$ **(c)** $\frac{5}{8}$ **(d)** $\frac{4}{9}$

27. The least number which when divided by 4, 6 and 10 leaves remainder 2 each time is: 1

(a) 122 **(b)** 58 **(c)** 60 **(d)** 62

28. The value of k for which $x = 3$ is a zero of $x^2 + kx - 6$ is: 1

(a) -1 **(b)** 1 **(c)** 5 **(d)** -3

29. For the pair $kx + 3y = 4$ and $8x + 2y = 5$ to have a unique solution, k must NOT equal: 1

(a) 12 **(b)** $\frac{32}{5}$ **(c)** 11 **(d)** $\frac{16}{3}$

30. If $x = 5$ is a root of $x^2 - 11x + k = 0$, then $k =$ 1

(a) 80 **(b)** -30 **(c)** 30 **(d)** 6

31. The sum of the first 20 terms of the AP $-6, -2, 2, 6, \ldots$ is: 1

(a) 714 **(b)** 640 **(c)** 570 **(d)** 1280

32. A pole 8 m high casts a shadow 5 m long. A tower casting a shadow 40 m has height (m): 1

(a) 64 **(b)** 320 **(c)** 35 **(d)** 43

33. A circle has centre $(0, 0)$ and passes through $(5, 12)$. Its radius is: 1

(a) 7 **(b)** 17 **(c)** 169 **(d)** 13

34. $\dfrac{\sin 35^\circ}{\cos 55^\circ} + \dfrac{\cos 40^\circ}{\sin 50^\circ} =$ 1

(a) 0 **(b)** 1 **(c)** 4 **(d)** 2

35. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 12 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. The height of the tower is: 1

(a) 6 m **(b)** $4\sqrt{3}$ m **(c)** $6\sqrt{3}$ m **(d)** $12\sqrt{3}$ m

36. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 28 cm. The length of each tangent is 21 cm. The perimeter of the quadrilateral $PAOB$ is: 1

(a) 98 cm **(b)** 133 cm **(c)** 49 cm **(d)** 42 cm

37. The largest possible circle is cut from a square sheet of side 70 cm. The area of the sheet left over is: 1

(a) 2100 cm² **(b)** 4900 cm² **(c)** 1050 cm² **(d)** 3850 cm²

38. A hemispherical bowl has outer radius 8 cm and inner radius 2 cm. The volume of material used to make the bowl is: 1

(a) 528 cm³ **(b)** 1056 cm³ **(c)** 2112 cm³ **(d)** 3168 cm³

39. For the data with classes 70–80, 80–90, 90–100, 100–110, 110–120, 120–130 with frequencies 5, 13, 5, 2, 4, 11 respectively, The mode of the data is: 1

(a) 85 **(b)** 90 **(c)** 80 **(d)** 100

40. 40 cards numbered 1 to 40 are placed in a box, mixed well and one is drawn at random. The probability that the number drawn is a composite number is: 1

(a) $\frac{13}{20}$ **(b)** $\frac{27}{40}$ **(c)** $\frac{13}{40}$ **(d)** $\frac{7}{10}$

41. The decimal expansion of $\frac{129}{312500}$ is: 1

(a) non-terminating repeating **(b)** non-terminating non-repeating

(c) terminating **(d)** an integer

42. If the zeroes of $x^2 + px + q$ are 2 and 7, then p equals: 1

(a) 9 **(b)** -14 **(c)** -9 **(d)** 14

43. If $4x + 3y = 31$ and $3x + 4y = 25$, then $x + y =$ 1

(a) 56 **(b)** 8 **(c)** 28 **(d)** 6

44. The equation $x^2 + 4x + 4 = 0$ has: 1

(a) two equal real roots **(b)** three real roots

(c) no real roots **(d)** two distinct real roots

45. The middle term of an AP with 11 terms, first term 8 and last term 48, is: 1

(a) 28 **(b)** 32 **(c)** 24 **(d)** 48

46. The diagonals of a rhombus are 10 and 24. Its side squared is: 1

(a) 289 **(b)** 676 **(c)** 338 **(d)** 169

47. The ratio $k : 1$ in which the line $x = 4$ divides the join of $(1, 0)$ and $(10, 0)$ has $k =$ 1

(a) 2 **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{3}$

48. If $\tan\theta = \frac{12}{5}$, then $\dfrac{\sin\theta + \cos\theta}{\cos\theta} =$ 1

(a) $\frac{12}{5}$ **(b)** $\frac{5}{17}$ **(c)** $\frac{7}{5}$ **(d)** $\frac{17}{5}$

49. Two points on level ground lie on opposite sides of a tower of height $12\sqrt{3}$ m, in line with its foot. The angles of elevation of the top of the tower from the two points are 30° and 60°. The distance between the two points is: 1

(a) 24 m **(b)** 96 m **(c)** $48\sqrt{3}$ m **(d)** 48 m

50. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 36 cm. The length of each tangent is 27 cm. The area of the quadrilateral $PAOB$ (in cm^2) is: 1

(a) 126 cm² **(b)** 972 cm² **(c)** 1944 cm² **(d)** 486 cm²

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND